Engineering tolerance analysis report

Preliminary Case Study 01 - Cover-Recess - Chain B - Conditional cold right gap

Dimensional units: mm | Application version: 0.1.0

Methodology: deterministic linear 1D worst-case analysis.

Issue time (UTC): Not supplied (deterministic export)

Project input SHA-256: 5ef3ea8afb2ed539ff397c0b40da43e703fa09fcf071bfce1a1af159b973427f

1. Functional assessment

Acceptance: Not assessed

No functional acceptance assessment was performed; functional limits were not provided.

Quantity	Exact value (mm)
Minimum result	0.350
Nominal result	0.500
Maximum result	0.710
Functional lower limit (inclusive)	Not provided
Functional upper limit (inclusive)	Not provided

Signed range: Entirely positive

Sign follows the chosen reference direction. Its physical meaning depends on the chain definition. For a clearance chain, crossing zero may indicate a transition between clearance and interference; interference is not guaranteed.

Schematic interval overview. Numerical tables contain the exact values; plot coordinates do not influence acceptance.

2. Complete dimensional inputs

All dimensional values in mm. Deviations are signed offsets from nominal. Contribution sign is separate from the nonnegative physical length. Blank dimension cells on continued rows carry the preceding identifier.

Dimension	Label	Nominal	Lower deviation	Upper deviation	Sign
H	H_L to H_R - proposed housing stop separation at 20 C	14.700	-0.100	0.100	+1
W	R_L to R_R - outer-ring width; selected open SKF 6204 Explorer, October 2018 catalogue; current procurement unverified	14.000	-0.060	0.000	-1
P	H_L to R_L - assumed cold left gap; not measured or guaranteed	0.200	-0.050	0.050	-1

3. Auditable contribution trace

Physical limits and signed chain contributions are distinct. All values in mm. A positive contribution uses physical minimum/maximum directly. A negative contribution subtracts physical maximum for its minimum contribution and physical minimum for its maximum. Signed nominal contributions apply the sign to nominal. Totals below are copied from the engine.

Dimension	Physical minimum	Physical maximum	Signed minimum	Signed nominal	Signed maximum
H	14.600	14.800	14.600	14.700	14.800
W	13.940	14.000	-14.000	-14.000	-13.940
P	0.150	0.250	-0.250	-0.200	-0.150
Total	-	-	0.350	0.500	0.710

4. Engineering assumptions and limitations

1. Linear one-dimensional additive chain only; contributions have fixed signs +1 or -1.

2. All dimensions and functional limits use the declared project unit. No unit conversion is performed.

3. Dimensional limits are independently and simultaneously attainable. Correlated/shared dimensions require engineering review before this model is used.

4. Functional acceptance requires full interval containment, including boundary equality. There is no acceptance rounding, hidden epsilon, or clamping of negative results.

5. Worst-case bounds are deterministic limits, not statistical probabilities or yield estimates.

6. This model excludes GD&T, multi-axis or nonlinear geometry, thermal effects, and deformation.

7. No ISO/ASME certification or standards compliance is claimed. Software checks do not establish physical suitability or safety. The engineer must verify the chain, reference direction, and limits.

8. Exact Decimal text is retained, including trailing zeros and scientific notation. Long values and labels wrap across lines/pages without truncation. Table headers repeat on continued pages.